

Getting Started with Rust, the Friendly Systems Programming Language

From Mozilla Research comes this safe, concurrent, practical systems programming language called Rust. It has been ranked the 'Most loved programming language' in the Stack Overflow Developer Survey for the past three years in succession.



Rust is the new, upcoming systems programming language that is experiencing tremendous popularity because of its features, efficiency and user-friendliness. It is quite similar in syntax to simple programming languages. The compilations are also in the form of bash Linux commands, so Rust can be used by most Linux users with ease.

Installation

Rust is a pretty easy to install. It can work on almost all platforms such as Linux, Mac and Windows, including on x86 bit systems and x64 bit systems.

For installing it in Linux, a simple shell script command (which includes the website location for downloading) using the 'curl' command does the trick. For Windows and Mac, we can download the setup file directly from the official website and run it easily on the respective platforms.

Rust is open source and, hence, there are no specifications or rules for using this language.

Special features of Rust

Rust has a number of important features that help to overcome the issues with many other programming languages and, therefore, increase both the safety and performance of a program. It offers features such as moving semantics, which means that we can modify the semantics as per our needs. It is a neutral form of language and can be moulded in any form, unlike other languages. Rust has generics that are based on traits; therefore, it can forward the hierarchy with the help of inheritance. It can also bind with other languages such as C with great efficiency and ease. Rust also provides the feature of pattern matching, which helps us to sequence-check algorithmic processes. Its other features include threads without data race conditions, zero-cost abstraction, minimal runtime and type inference.